

Supplementary material

Recognising the phone, not the throat: deconfounded evaluation and shortcut mitigation for smartphone pharyngitis classification

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S1 Differences between the two phones

Table S1 compares the symptom records of patients photographed with each phone (chi-square tests, Bonferroni correction over 20 symptoms). Samsung patients reported more symptoms (mean 2.7 vs 1.7, Mann–Whitney $p < 10^{-19}$), mostly systemic ones, yet had a lower bacterial prevalence. Labelling also differed (main text, Table 1): Samsung patients were rated by more physicians (median 5 vs 4; only Samsung patients had 8 or 9 raters), physicians agreed more often (mean agreement 0.66 vs 0.47, $p < 10^{-11}$), and ties were rarer (6.3% vs 23.1%, $p < 10^{-9}$).

S2 Validation of the deconfounded AUC: full results

Table S2 gives the simulation results summarised in main-text Fig. 3A (500 replicates per setting, 700 patients, two sites, overall prevalence 0.28). The *target* is the AUC of the same score in a large unconfounded population from the same sites (prevalence gap 0). The deconfounded AUC was unbiased for this target in every setting (largest absolute bias 0.001). The adjusted AUC was unbiased for the marker-only and site-only scores but overestimated the target of the learned model by up to 0.024, because within-site comparisons ignore the site offset that the learned model adds to its scores. Table S3 gives the resampling results on PGUPharyngitis summarised in main-text Fig. 3B (20 resamples per gap, 200 patients per phone, overall prevalence 0.25).

S3 Mitigation: adjusted AUC and cross-phone transfer

Table S4 adds the covariate-adjusted AUC [1] to main-text Table 5. No mitigation method changed the adjusted AUC significantly (corrected t -test, all $p > 0.10$). Table S5 gives cross-phone transfer

Table S1: Share of patients with each symptom, by phone (%), ordered by p value.

Symptom	Samsung ($n = 382$)	Xiaomi ($n = 290$)	p	p (Bonferroni)
Myalgia	26.2	10.0	2.3 e−07	4.6 e−06
Cephalalgia	21.7	7.6	9.9 e−07	2 e−05
Lethargy	12.8	3.1	1.7 e−05	0.00033
Diarrhea	6.3	0.3	0.00013	0.0026
Pyrexia	25.4	13.4	0.0002	0.004
Vomit	11.3	3.8	0.00072	0.014
Otalgia	7.6	1.7	0.0011	0.022
Vertigo	6.8	1.4	0.0014	0.029
Rhinorrhea	32.2	21.7	0.0035	0.07
Cough	28.5	39.0	0.0057	0.11
Sputum	12.0	5.9	0.0096	0.19
Rigors	9.9	4.5	0.012	0.25
Abdominal Pain	4.5	1.0	0.019	0.37
Sternutation	3.9	1.0	0.04	0.79
Sore Throat	47.1	40.7	0.11	1
Nasal Congestion	2.1	0.7	0.24	1
Anorexia	2.1	1.0	0.44	1
Eye Infection	2.1	1.0	0.44	1
Feeling Cold	2.9	4.1	0.5	1
Asthma	2.9	3.4	0.84	1

Table S2: Simulation. Mean metric (mean bias against the target) over 500 replicates per setting; last column: SD of the deconfounded / adjusted AUC across replicates.

Score	Gap	Target	Pooled	Deconfounded	Adjusted	SD
Marker only	0.0	0.600	0.600 (0.000)	0.600 (0.000)	0.600 (0.000)	0.023 / 0.023
	0.1	0.600	0.600 (0.000)	0.600 (0.000)	0.600 (0.000)	0.025 / 0.025
	0.2	0.600	0.601 (+0.001)	0.601 (+0.001)	0.600 (0.000)	0.025 / 0.025
	0.3	0.600	0.600 (0.000)	0.601 (0.000)	0.601 (+0.001)	0.025 / 0.025
	0.4	0.600	0.601 (+0.001)	0.601 (+0.001)	0.601 (+0.001)	0.031 / 0.027
Site only	0.0	0.500	0.501 (+0.001)	0.501 (+0.001)	0.502 (+0.002)	0.015 / 0.026
	0.1	0.500	0.560 (+0.060)	0.500 (0.000)	0.501 (+0.001)	0.014 / 0.024
	0.2	0.500	0.618 (+0.118)	0.500 (0.000)	0.501 (+0.001)	0.016 / 0.027
	0.3	0.500	0.678 (+0.178)	0.499 (-0.001)	0.499 (-0.001)	0.015 / 0.025
	0.4	0.500	0.737 (+0.237)	0.501 (+0.001)	0.502 (+0.002)	0.017 / 0.027
Learned (marker + site)	0.0	0.597	0.597 (0.000)	0.597 (0.000)	0.599 (+0.002)	0.023 / 0.023
	0.1	0.584	0.613 (+0.029)	0.584 (0.000)	0.594 (+0.010)	0.025 / 0.025
	0.2	0.564	0.651 (+0.087)	0.564 (0.000)	0.583 (+0.019)	0.023 / 0.026
	0.3	0.548	0.701 (+0.153)	0.547 (0.000)	0.570 (+0.022)	0.021 / 0.027
	0.4	0.535	0.754 (+0.219)	0.536 (+0.001)	0.559 (+0.024)	0.023 / 0.029

Table S3: PGUPharyngitis resampled to a given prevalence gap between phones. Mean (SD) over 20 resamples, each evaluated with stratified 5-fold cross-validation. At gap 0 the phone carries no label information.

Model	Gap	Pooled AUC	Deconfounded AUC	Adjusted AUC
Phone only	0.0	0.500 (0.000)	0.500 (0.000)	0.500 (0.000)
	0.1	0.567 (0.000)	0.500 (0.000)	0.500 (0.000)
	0.2	0.633 (0.000)	0.500 (0.000)	0.500 (0.000)
	0.3	0.700 (0.000)	0.500 (0.000)	0.500 (0.000)
	0.4	0.767 (0.000)	0.500 (0.000)	0.500 (0.000)
Image (DINOv2)	0.0	0.535 (0.032)	0.535 (0.032)	0.534 (0.033)
	0.1	0.579 (0.028)	0.546 (0.027)	0.554 (0.035)
	0.2	0.654 (0.023)	0.552 (0.027)	0.578 (0.035)
	0.3	0.714 (0.021)	0.539 (0.025)	0.572 (0.034)
	0.4	0.775 (0.014)	0.533 (0.028)	0.567 (0.031)
Image after LEACE	0.0	0.537 (0.032)	0.537 (0.032)	0.536 (0.032)
	0.1	0.546 (0.035)	0.541 (0.035)	0.543 (0.036)
	0.2	0.567 (0.037)	0.554 (0.036)	0.560 (0.038)
	0.3	0.557 (0.043)	0.538 (0.051)	0.548 (0.046)
	0.4	0.559 (0.039)	0.541 (0.056)	0.547 (0.047)
Symptoms	0.0	0.528 (0.036)	0.528 (0.036)	0.528 (0.037)
	0.1	0.543 (0.037)	0.537 (0.038)	0.533 (0.036)
	0.2	0.543 (0.034)	0.517 (0.042)	0.517 (0.033)
	0.3	0.576 (0.037)	0.534 (0.054)	0.529 (0.037)
	0.4	0.587 (0.027)	0.499 (0.053)	0.508 (0.031)

with colour-constancy features and per-phone standardisation (each phone standardised with its own statistics, which needs no labels from the test phone).

Table S4: Pooled, deconfounded and adjusted AUC of the mitigation methods (frozen embeddings, 5×5 cross-validation, mean over repeats).

Method	DINOv2			ConvNeXt-Tiny		
	Pooled	Deconf.	Adjusted	Pooled	Deconf.	Adjusted
None (reference)	0.700	0.558	0.599	0.681	0.545	0.563
Colour constancy	0.698	0.549	0.594	0.672	0.542	0.554
Phone as covariate	0.699	0.558	0.598	0.679	0.546	0.562
Reweighting	0.631	0.564	0.577	0.565	0.518	0.522
Per-phone standardisation	0.583	0.577	0.589	0.543	0.547	0.551
LEACE	0.579	0.559	0.571	0.546	0.533	0.543
LEACE + reweighting	0.574	0.553	0.569	0.545	0.532	0.544
Colour + LEACE + reweighting	0.579	0.556	0.572	0.535	0.524	0.530
Network, no adversary	0.662	0.560	0.569	0.653	0.566	0.585
Adversarial, $\lambda = 0.3$	0.652	0.560	0.569	0.630	0.559	0.574
Adversarial, $\lambda = 1$	0.638	0.556	0.564	0.612	0.555	0.564
Adversarial, $\lambda = 3$	0.625	0.549	0.553	0.607	0.555	0.564
Adversarial, $\lambda = 10$	0.626	0.546	0.556	0.588	0.557	0.559

S4 Gated cross-modal fusion network

The image embedding passes through layer normalisation, dropout and a linear layer to a 128-dimensional vector h_{img} ; the 22 tabular inputs pass through a 22–64–128 perceptron to h_{sym} . A gate computes a scalar weight per patient,

$$g = \sigma(W[h_{\text{img}}; h_{\text{sym}}] + b), \quad z = g h_{\text{img}} + (1 - g) h_{\text{sym}}, \quad (\text{S1})$$

Table S5: Cross-phone transfer (AUC [95% bootstrap CI]): trained on all patients of one phone, tested on the other.

Features	Correction	Samsung $\rightarrow$ Xiaomi	Xiaomi $\rightarrow$ Samsung
DINOv2	None	0.519 [0.45, 0.59]	0.502 [0.42, 0.58]
	Colour constancy	0.537 [0.47, 0.60]	0.535 [0.45, 0.62]
	Per-phone standardisation	0.528 [0.46, 0.59]	0.508 [0.43, 0.59]
	Colour + standardisation	0.538 [0.47, 0.60]	0.536 [0.46, 0.61]
ConvNeXt-Tiny	None	0.530 [0.46, 0.60]	0.507 [0.42, 0.59]
	Colour constancy	0.596 [0.53, 0.66]	0.569 [0.49, 0.64]
	Per-phone standardisation	0.528 [0.46, 0.59]	0.508 [0.43, 0.59]
	Colour + standardisation	0.594 [0.53, 0.66]	0.575 [0.50, 0.65]

and a dropout-linear head maps z to a logit. Auxiliary image-only and symptom-only heads add their losses with weight λ , and modality dropout (probability 0.2 per branch, never both) randomly removes one branch during training. Training used the soft label (share of physicians voting bacterial) as target in class-weighted binary cross-entropy, AdamW with weight decay 10^{-4} , batch 32 and a cosine schedule over 60 epochs. Each epoch drew one of the ten cached augmentation embeddings per patient. The inner cross-validation selected the learning rate $\{10^{-3}, 3 \times 10^{-4}\}$, dropout $\{0.3, 0.5\}$, $\lambda \in \{0, 0.3\}$ and the number of epochs (maximum of the smoothed mean inner AUC curve); the model was then retrained on the full training part. Ablations replaced the gate by concatenation or by FiLM [2], used hard labels, removed the auxiliary heads, removed modality dropout, or removed fever and cough from the inputs.

Ablations No component made a measurable difference (Fig. S1). With ConvNeXt features, the full gated model reached 0.684; replacing the gate by concatenation gave 0.686 ($p = 0.60$), FiLM 0.681, removing the auxiliary heads 0.682 and removing modality dropout 0.682. Training on hard instead of soft labels lowered AUC to 0.672 (difference 0.012, $p = 0.19$). Removing fever and cough did not hurt (0.690). A shuffled-image control, in which image embeddings were permuted across patients, fell to 0.529, below symptoms alone, so the pipeline does not leak labels.

Gate The gate weighted the image at $g \approx 0.48$ on average (DINOv2) and slightly less when physicians disagreed (median 0.47 vs 0.50, Mann-Whitney $p < 0.001$; Fig. S2).

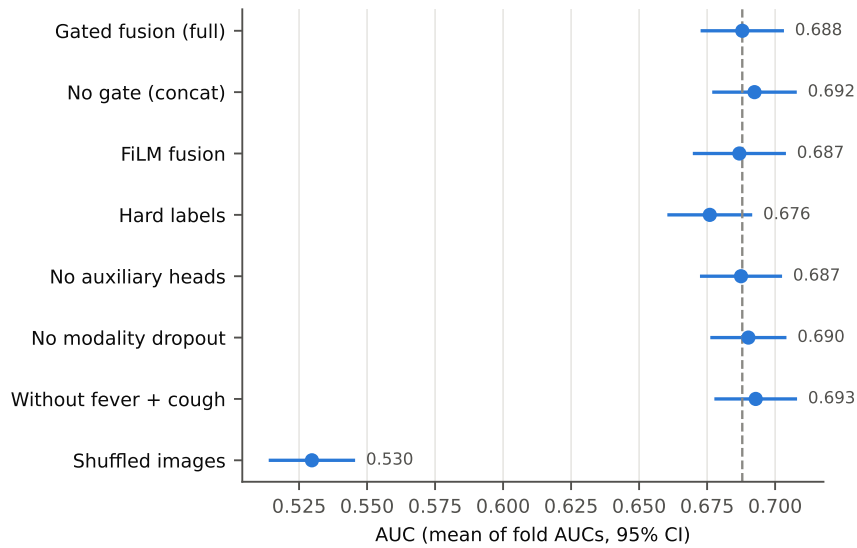


Figure S1: Ablation of the gated fusion network (ConvNeXt features): mean fold AUC with 95% confidence interval. The shuffled-image control falls to chance.

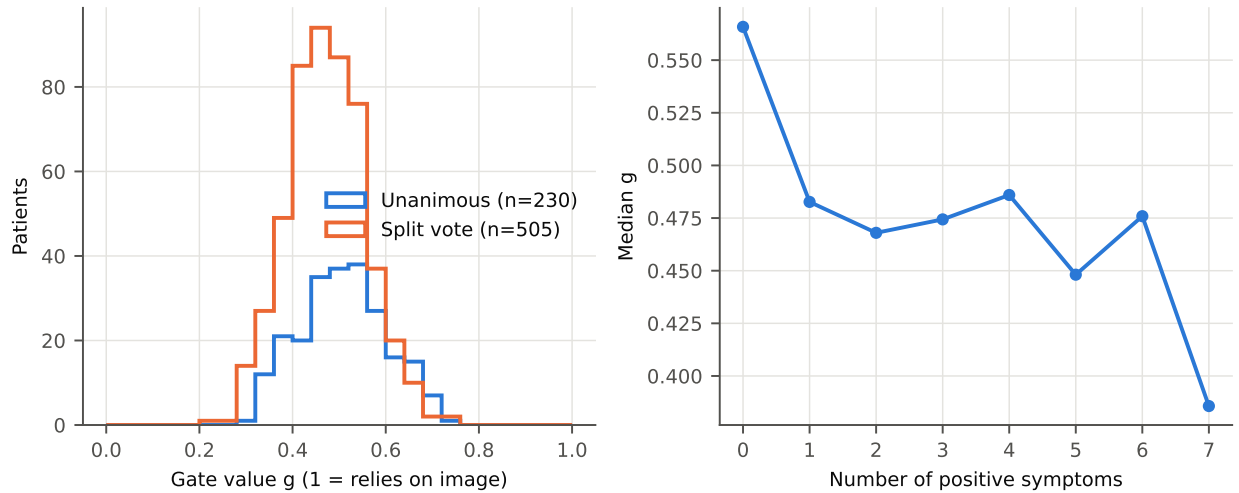


Figure S2: Gate values of the gated fusion model (DINOv2): distribution for unanimous and split votes (left) and median gate value by number of positive symptoms (right). $g = 1$ means the model relies only on the image.

S5 Planned fusion hypotheses under three tie rules

Gated fusion (DINOv2) beat the best symptom model by 0.095 AUC (Holm-adjusted $p = 0.012$; significant in 5 of 5 DeLong repeats) but did not beat the best image-only model (-0.020 , $p = 1.0$) or the best standard fusion (-0.008 , $p = 1.0$). Under the two alternative tie rules the advantage over symptoms was no longer significant (Table S6).

Table S6: Planned comparisons for gated fusion (DINOv2). Δ AUC is the mean fold-level difference; p values are one-sided, Nadeau–Bengio corrected and Holm-adjusted.

Comparison	Ties bacterial	Ties non-bacterial	Ties dropped
H1: vs best symptom-only	+0.095 ($p = 0.012$)	+0.044 ($p = 0.38$)	+0.051 ($p = 0.22$)
H2: vs best image-only	-0.020 ($p = 1.0$)	-0.009 ($p = 1.0$)	-0.002 ($p = 1.0$)
H3: vs best standard fusion	-0.008 ($p = 1.0$)	-0.018 ($p = 1.0$)	-0.019 ($p = 1.0$)
Image (DINOv2): AUC / within phone	0.700 / 0.586	0.671 / 0.609	0.696 / 0.605
Phone only: AUC	0.673	0.618	0.650
Best symptom-only: AUC	0.596	0.616	0.639

S6 ROC curves, calibration and clinical utility

Among the frozen-feature image and fusion models, Brier scores ranged from 0.188 (gated fusion, DINOv2) to 0.226 (stacking, ConvNeXt); the phone-only model scored 0.182. Class-balanced training made image and fusion models overestimate risk, and decision-curve analysis [3] showed no net benefit over treating everyone at thresholds up to 0.20; at 0.30 the stacked model gave a net benefit of 0.024 versus -0.030 for treat-all, but the phone-only model gave 0.082 (Fig. S4).

S7 Additional shortcut controls and label dependence

Large images Restricting to photographs with an original short side of at least 500 pixels did not change within-phone AUC (DINOv2: 0.568 Samsung, 0.611 Xiaomi), so very small crops do not explain the shortcut.

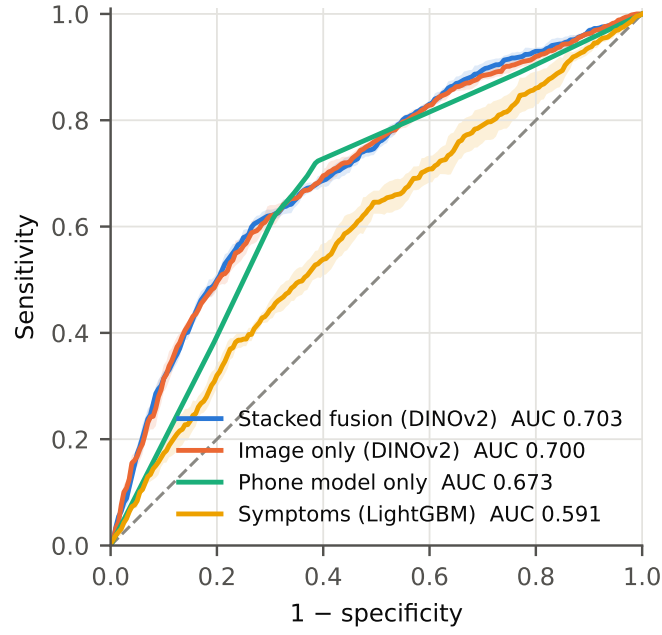


Figure S3: Mean ROC curves (± 1 SD over five repeats) for stacked fusion (DINOv2), the frozen DINOv2 image model, the phone-only model and the LightGBM symptom model. Dashed line: chance.

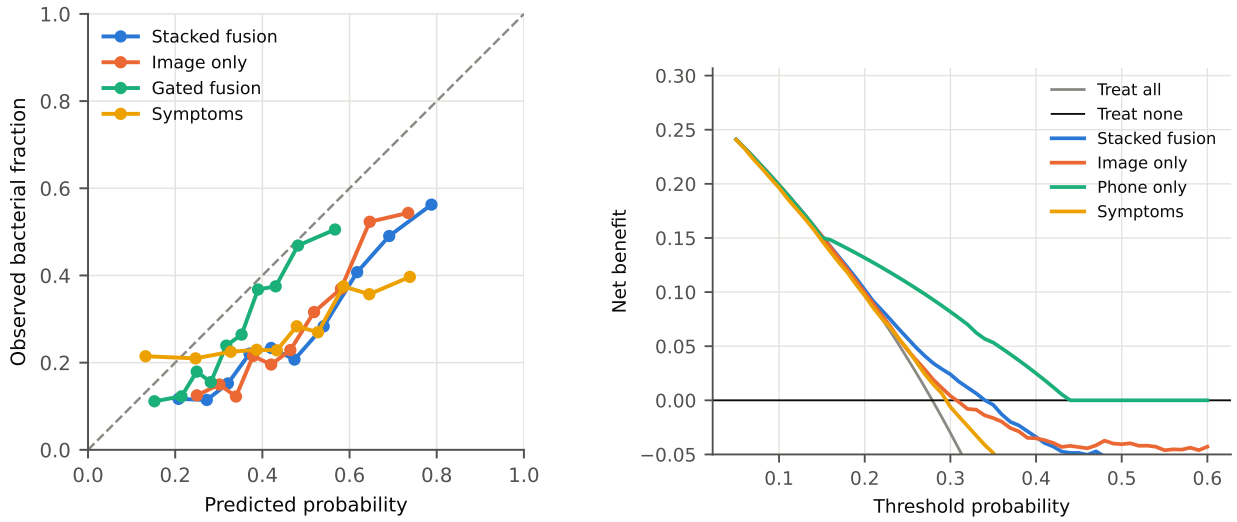


Figure S4: (A) Calibration curves and (B) decision curves for the main models. Treat-all and treat-none are shown for reference.

Label dependence Models discriminated better on unanimous than on split votes (DINOv2 image: 0.719 vs 0.678; symptoms: 0.623 vs 0.580). Image predictions correlated more with the physicians’ vote share than symptom predictions did (Spearman 0.32 vs 0.13), though the phone-only model also reached 0.28. In a residual test, DINOv2 embeddings predicted the out-of-fold errors of the symptom model (ridge regression, 5-fold cross-validated $r = 0.29$; permutation $p = 0.002$ with 500 permutations), but this test cannot separate throat appearance from phone identity.

S8 Explainability and subgroups

We computed SHAP values [4] for a LightGBM symptom model and Grad-CAM maps [5] from a ConvNeXt-Tiny fine-tuned on all patients for the median best epoch count (seven), used only for visualisation. Cough, age, vomiting, myalgia and sore throat had the largest mean absolute SHAP values (Fig. S5). Grad-CAM maps of the most confident bacterial cases highlighted white spots in close-up photographs, whereas the most confident non-bacterial cases were wide views of the oral cavity with the uvula and tongue visible (Fig. S6); framing and zoom may therefore act as an additional site cue. The image model discriminated better in children aged 14 or younger (AUC 0.77) than in adults aged 45 or older (0.64), and similarly in women (0.68) and men (0.72) (Fig. S7).

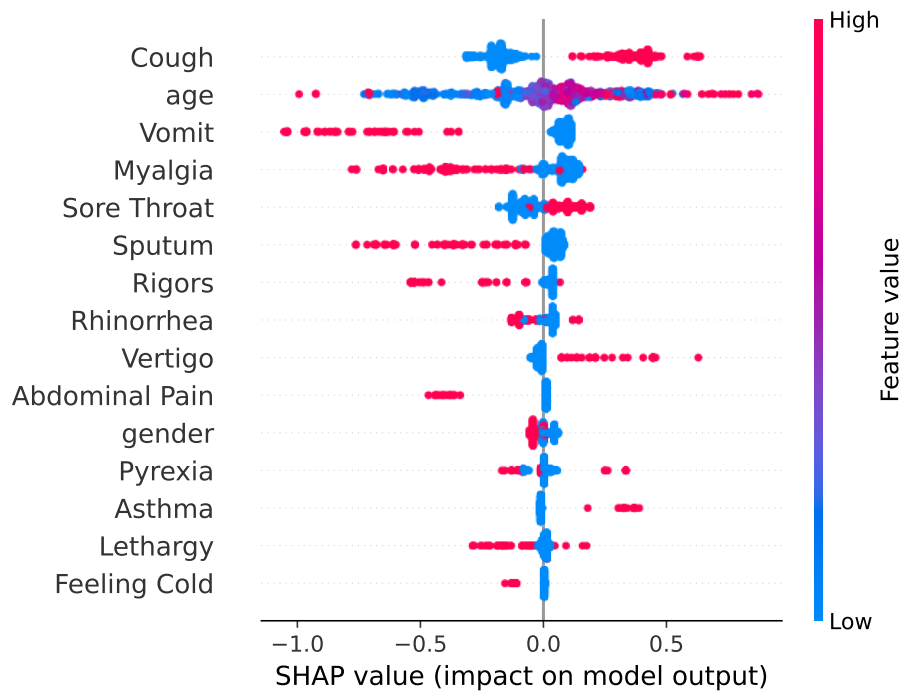


Figure S5: SHAP summary of the LightGBM symptom model fitted on all patients.

References

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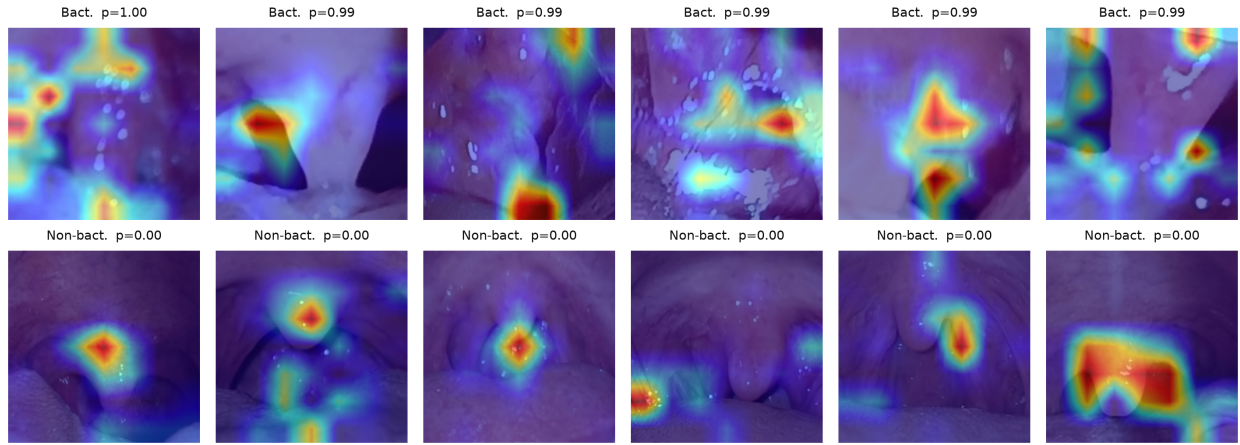


Figure S6: Grad-CAM maps of a ConvNeXt-Tiny fine-tuned on all patients for the six most confident bacterial (top) and non-bacterial (bottom) cases. Red marks high attention. Bacterial predictions are close-ups; non-bacterial predictions are wide views.

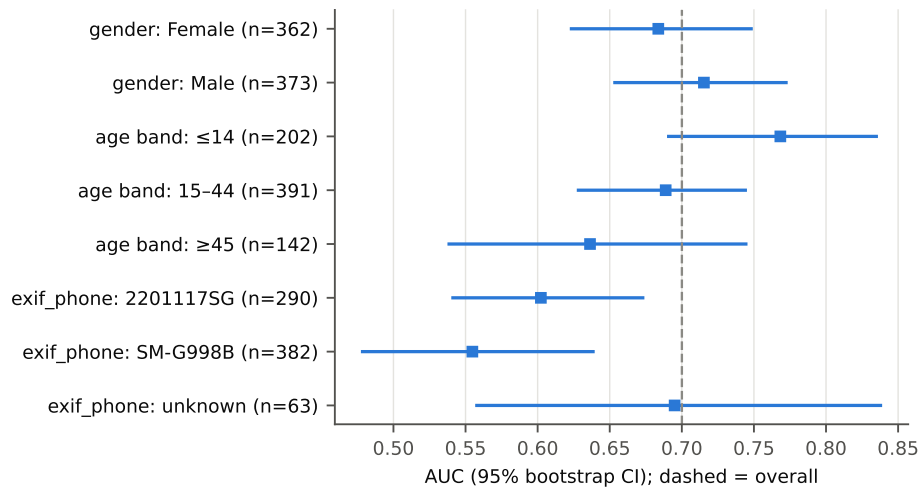


Figure S7: AUC of the frozen DINOv2 image model by sex, age band and phone, with 95% bootstrap confidence intervals. Dashed line: overall AUC.

- [4] Scott M. Lundberg and Su-In Lee. A unified approach to interpreting model predictions. In *Advances in Neural Information Processing Systems*, volume 30, 2017.
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